# 結合單點感測與住戶回饋的可追溯居家環境與外出建議代理人


## 摘要

住戶決定何時除濕、開窗或外出，須同時考量房間環境與目的地天氣。公開佔用資料以房間是否有人為標記，缺乏辨識冷氣、除濕與開窗所需的操作標記。本研究聚焦單一房間，整合 ESP32 溫濕度量測、中央氣象署 368 鄉鎮預報快照與住戶回饋，提出可追溯的生活建議代理人。代理人透過限定工具取得證據，回覆附資料來源、適用地點、時段及規則版本。系統合併候選事件，按設定間隔請住戶確認，並區分模型推論、住戶確認與未知狀態。研究規劃蒐集約兩週的量測、操作日誌及回覆，按時間保留較晚的完整事件，在同一測試集比較固定規則與決策樹。詢問策略以同批事件回放比較，指標包括每小時詢問數、回覆率及重複詢問率。另以共同情境評估回答的證據一致率。未知標記獨立計數，樣本不足時呈現探索性結果。預期交付操作標記資料、建議原型及可重現的評估流程，以界定單點感測的操作辨識範圍。季節變異留待後續研究。



**關鍵詞** 主動詢問、操作事件辨識、物聯網、住戶回饋、住宅微氣候、可追溯建議、氣象預報整合


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# A Traceable Recommendation Agent for Indoor Comfort and Outdoor Activities Using Single-Point Sensing and Occupant Feedback


**Abstract—** Decisions about when to dehumidify a room, open a window, or go outdoors depend on conditions in the room and weather at the destination. Public occupancy datasets label human presence. Identifying air-conditioning, dehumidification, and window-opening events requires labels for the operations themselves. This study proposes a single-room recommendation agent that integrates ESP32-based temperature and relative humidity measurements, timestamped forecast snapshots for 368 township-level areas from Taiwan's Central Weather Administration, and occupant feedback. The agent retrieves evidence through a restricted set of tools and links each recommendation to its data sources, applicable location and time window, and rule version. The system merges candidate events, spaces confirmation requests, and records model inferences, occupant confirmations, and unknown states separately. The planned evaluation will use approximately two weeks of measurements, operation logs, and occupant responses. Fixed rules and a decision tree will be compared on the same test set of complete events held out chronologically. Prompting strategies will be compared by replaying the same events and measuring prompts per hour, response rate, and repeated-prompt rate. Recommendation consistency with supporting evidence will be assessed on a common set of scenarios. Unknown labels will be counted separately, and findings based on limited samples will be reported as exploratory. Expected contributions include a dataset with confirmed operation labels, a traceable recommendation prototype, and a reproducible protocol for determining which operations can be distinguished using single-point sensing. Seasonal variability remains a subject for future study.



**Index Terms—** Active querying, event recognition, Internet of Things, occupant feedback, residential microclimate, traceable recommendations, weather forecast integration.